Package ‘scScale’

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Title Single-Cell Scaling Law Utilities

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Description Median Marchenko-Pastur noise calibration, BBP spike inversion, spectral overlap, and task-difficulty utilities for single-cell scaling analyses, aligned with ChesleaZ/scScale.

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URL https://github.com/ChesleaZ/scScale

BugReports https://github.com/ChesleaZ/scScale/issues

Encoding UTF-8

Depends R (>= 2.10),
Matrix

LazyData true

LazyDataCompression xz

Suggests irlba,
SeuratObject

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.2

Contents

`gse123025_myeloid_hvg` *GSE123025 Myeloid HVG Count Matrix Example*

Description

A larger example subset derived from GSE123025 for RMT spike-model tutorials. The object keeps all cells from the source matrix and restricts genes to 2,000 highly variable genes selected by `select_hvgs()`.

Usage

```
data(gse123025_myeloid_hvg)
```

Format

A list with:

counts Integer matrix with 2,000 selected genes/features by 1,922 cells.

source Source file name in the canonical lab data directory.

accession GEO accession identifier, `"GSE123025"`.

description Short description of the HVG subset.

selection List describing the feature and cell selection.

original_dim Dimensions of the source matrix before HVG selection.

cell_totals Column sums of the HVG count matrix.

gene_totals Row sums of the HVG count matrix.

Source

Derived from `data/GSE123025/GSE123025_Single_myeloid_1922_cells_processed_data.csv.gz` in the canonical lab workspace. The source matrix has 23,429 genes by 1,922 cells and is about 8.0 MB compressed as CSV gzip, or 95 MB uncompressed.

Examples

```
data(gse123025_myeloid_hvg)
counts <- gse123025_myeloid_hvg$counts
dim(counts)
fit <- scscale_fit(counts, r = 10, mp_max_iter = 3, fit_umi = FALSE)
fit
```

gse164378_3p_citeseq_hvg

GSE164378 3 Prime CITE-seq RNA HVG and ADT Example

Description

A compact paired RNA and ADT example derived from the 3 prime CITE-seq portion of GSE164378. The object keeps 4,000 matched PBMC cells, 2,000 RNA highly variable genes, and all 228 measured ADT features.

Usage

```
data(gse164378_3p_citeseq_hvg)
```

Format

A list with:

rna_counts Sparse integer matrix with 2,000 selected RNA genes by 4,000 cells.

adt_counts Sparse integer matrix with 228 ADT features by the same 4,000 cells.

meta Cell metadata, including RNA/ADT QC totals, lane, donor, time, batch, and cell type labels.

source Short source dataset description.

accession GEO accession identifier, `"GSE164378"`.

description Short description of the compact paired-modality subset.

selection List describing the cell and RNA feature selection.

original_dim Dimensions of the source 3 prime RNA and ADT matrices before subsetting.

cell_totals RNA and ADT column sums for the selected cells.

feature_totals RNA and ADT row sums for the selected features.

Source

Derived from the 3 prime RNA and ADT matrices in GSE164378, the Hao et al. multimodal PBMC reference dataset. The source 3 prime matrices have 33,538 RNA features by 161,764 cells and 228 ADT features by the same 161,764 cells.

Examples

```
data(gse164378_3p_citeseq_hvg)
rna <- gse164378_3p_citeseq_hvg$rna_counts
adt <- gse164378_3p_citeseq_hvg$adt_counts
dim(rna)
dim(adt)
```

scScale *Single-Cell Scaling Law Utilities*

Description

Gaussian spike fitting and low-rank spectral mutual information utilities for single-cell scaling-law analyses.

Details

The recommended paired-modality workflow is:

1. Fit matched modalities once with `scscale_pair_fit`.
2. Fit UMI-depth scaling with `scscale_umi_mi`.
3. Fit cell-number scaling with `scscale_cell_number_mi`.
4. Fit the joint cell-number by UMI grid with `scscale_cell_number_by_umi_mi`.
5. Compare with direct empirical MI from `scscale_empirical_mi`.

All theory curves are routed through `scscale_low_rank_mi`, the single low-rank subspace-alignment MI formula.

See Also

`scscale_fit`, `scscale_pair_fit`, `scscale_low_rank_mi`, `scscale_umi_mi`, `scscale_cell_number_mi`, `scscale_cell_number_by_umi_mi`, `scscale_empirical_mi`

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scscale_cell_number_by_umi_mi
Fit Joint Cell-Number by UMI MI Scaling

Description

Compute the theory MI grid over both cell number and UMI depth, reusing the refitted UMI spike strengths and subspace alignments.

Usage

```
scscale_cell_number_by_umi_mi(pair, umi, n_grid,
  sampling_rates = umi$sampling_rates)
```

Arguments

`pair`	A `scscale_pair_fit` object.
`umi`	A `scscale_umi_mi` object.
`n_grid`	Cell-number grid.
`sampling_rates`	Subset or ordering of UMI sampling rates to use.

Value

A data frame with `n`, `c_X`, `I_theory`, and `sampling_rate`.

scscale_cell_number_mi — *Fit Cell-Number MI Scaling*

Description

Compute the theory MI curve as the number of cells varies.

Usage

```
scscale_cell_number_mi(pair, n_grid, q_X = pair$x_fit$spikes$q_X,
  P = pair$P)
```

Arguments

`pair`	A `scscale_pair_fit` object.
`n_grid`	Cell-number grid.
`q_X`	Optional X-side spike-strength vector.
`P`	Optional X/Y low-rank subspace alignment matrix.

Value

A data frame with `n`, `c_X`, and `I_theory`.

scscale_empirical_mi — *Compute Empirical Subspace Mutual Information*

Description

Compute empirical low-rank subspace MI directly from data, either between two matrices or between one matrix and a cell-level target.

Usage

```
scscale_empirical_mi(x, target, r = 10, input = c("counts", "normalized"),
  target_input = input, target_depth = 1e4, count_transform = c("log1p_cpm",
  "pearson_residual", "log1p"), center = TRUE, scale = FALSE,
  eps = 1e-12, use_irlba = TRUE, store_subspaces = FALSE)
```

Arguments

`x`	Feature-by-cell matrix for the source modality.
`target`	Second feature-by-cell matrix or cell-level target vector.
`r`	Number of singular-vector directions to keep.
`input`, `target_input`	Input type for `x` and matrix targets.
`target_depth`, `count_transform`, `center`, `scale`	Preprocessing controls.

eps — Numerical clipping value.

use_irlba — Whether to use **irlba** for partial SVD when available.

store_subspaces — Whether to keep fitted subspace matrices in the output.

Value

A list containing empirical mi, canonical-correlation-style gamma, and effective rank information.

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scscale_fit *Fit a Gaussian Spike Model*

Usage

```
scscale_fit(x, input = c("counts", "normalized", "representation",
  "eigenvalues"), target = NULL, target_fit = NULL, n = NULL, p = NULL,
  target_depth = 1e4, count_transform = c("log1p_cpm",
  "pearson_residual", "log1p"), center = TRUE, scale = FALSE,
  n_features = NULL, min_cells = 10, mp_max_iter = 300, mp_grid_n = 20000,
  mp_stop_metric = c("qq_rmse_log", "ks", "neg_log_likelihood", "none"),
  mp_stop_tol = 1e-4, mp_min_iter = 2, r = NULL,
  fit_umi = input[1] == "counts",
  sampling_rates = c(0.10, 0.20, 0.35, 0.50, 0.70, 0.85, 1.00),
  U_grid = NULL, umi_seed = 1, umi_replicates = 1,
  store_matrix = FALSE, use_irlba = TRUE)
```

Arguments

x — Feature-by-cell counts, normalized matrix, representation matrix, or eigenvalue vector.

input — Input type.

target, target_fit — Optional target inputs used by legacy MI summaries.

n, p — Cell and feature counts, required when input = "eigenvalues".

target_depth, count_transform, center, scale, n_features, min_cells — Count preprocessing controls.

mp_max_iter, mp_grid_n, mp_stop_metric, mp_stop_tol, mp_min_iter — MAD/MP bulk and spike-fitting controls.

r — Optional number of spikes to keep.

fit_umi, sampling_rates, U_grid, umi_seed, umi_replicates — Legacy UMI scaling controls.

store_matrix — Whether to store the normalized matrix in the returned fit.

use_irlba — Whether to use **irlba** for partial SVD when available.

Value

An scscale_fit object containing the spectrum, MP bulk fit, spike table, recoverability vectors, and optional stored normalized matrix.

Examples

```
data(gse123025_myeloid_hvg)
x <- gse123025_myeloid_hvg$counts[1:100, 1:200]
fit <- scscale_fit(x, mp_grid_n = 600, mp_max_iter = 50)
head(fit$spikes)
```

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scscale_low_rank_mi — *Low-Rank Subspace Mutual Information*

Description

Compute the theory MI from X/Y recoverability and optional low-rank subspace alignment. This is the single theory formula used by the scaling helpers.

Usage

```
scscale_low_rank_mi(theta_X, theta_Y, P = NULL, eps = 1e-12)
```

Arguments

theta_X, theta_Y	Recoverability vectors for the X and Y low-rank signal directions.
P	Optional subspace alignment matrix between X and Y. If omitted, directions are matched by rank.
eps	Numerical clipping value.

Value

A list with mi, I_theory, singular values sigma, squared canonical correlations gamma, and effective ranks.

Examples

```
scscale_low_rank_mi(c(0.5, 0.2), c(0.4, 0.1))$mi
```

---

scscale_mi — *Compare Two Fitted scScale Objects*

Description

Compare two fitted objects with scscale_low_rank_mi and optional legacy scaling summaries.

Usage

```
scscale_mi(fit, target_fit, n_grid = NULL, U_grid = NULL,
  sampling_rates = NULL, combine = TRUE, empirical = TRUE,
  store_empirical_subspaces = FALSE, use_irlba = TRUE, r = NULL,
  P = NULL, eps = 1e-12)
```

Arguments

fit, target_fit	scscale_fit objects.
n_grid, U_grid, sampling_rates	Optional cell-number and UMI grids.
combine	Whether to return a combined legacy grid when both dimensions are supplied.
empirical	Whether to compute empirical subspace MI from stored matrices.
store_empirical_subspaces	Whether to keep empirical subspace matrices.
use_irlba	Whether to use **irlba** for partial SVD when available.
r	Optional number of directions to keep.
P	Optional low-rank subspace alignment matrix.
eps	Numerical clipping value.

Value

A list containing theory MI and optional empirical or scaling components.

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scscale_pair_fit *Fit Matched Modalities for MI Scaling*

Description

Fit both modalities, store their normalized matrices, compute right singular vectors, and align the low-rank subspaces.

Usage

```
scscale_pair_fit(x, y, count_transform = c("log1p_cpm",
  "pearson_residual", "log1p"), mp_max_iter = 300, mp_grid_n = 3000,
  use_irlba = TRUE, ...)
```

Arguments

x, y	Matched feature-by-cell count matrices for the two modalities.
count_transform	Count normalization transform passed to scscale_fit.
mp_max_iter, mp_grid_n	MAD/MP spike-fitting controls.
use_irlba	Whether to use **irlba** for partial SVD when available.
...	Additional arguments passed to scscale_fit.

Value

A scscale_pair_fit object with x_fit, y_fit, z_X, z_Y, the overlap matrix P, baseline mi, and I_infinity.

Examples

```
data(gse164378_3p_citeseq_hvg)
rna <- gse164378_3p_citeseq_hvg$rna_counts[1:100, 1:250]
adt <- gse164378_3p_citeseq_hvg$adt_counts[1:35, 1:250]
pair <- scscale_pair_fit(rna, adt, mp_grid_n = 600, mp_max_iter = 50)
pair$I_infinity
```

`scscale_umi_mi` *Fit UMI-Depth MI Scaling*

Description

Refit the X modality over UMI depth, recompute the X/Y subspace alignment, and return both the refitted theory MI curve and the refitted `I_infinity` bound.

Usage

```
scscale_umi_mi(pair, x_counts, sampling_rates = c(0.10, 0.20, 0.35,
  0.50, 0.70, 0.85, 1.00), count_transform = pair$count_transform,
  seed = 1, mp_max_iter = 300, mp_grid_n = 2000, use_irlba = TRUE)
```

Arguments

`pair`	A `scscale_pair_fit` object.
`x_counts`	Original X-side count matrix to downsample and refit.
`sampling_rates`	Global UMI sampling fractions in `(0, 1]`.
`count_transform`	Count transform used for refits.
`seed`	Random seed for downsampling.
`mp_max_iter, mp_grid_n`	MAD/MP spike-fitting controls for refits.
`use_irlba`	Whether to use **irlba** for partial SVD when available.

Value

A `scscale_umi_mi` object with `curve`, `q_by_rate`, `fit_by_rate`, and `P_by_rate`.

Examples

```
data(gse164378_3p_citeseq_hvg)
rna <- gse164378_3p_citeseq_hvg$rna_counts[1:100, 1:250]
adt <- gse164378_3p_citeseq_hvg$adt_counts[1:35, 1:250]
pair <- scscale_pair_fit(rna, adt, mp_grid_n = 600, mp_max_iter = 50)
umi <- scscale_umi_mi(pair, rna, sampling_rates = c(0.5, 1),
  mp_grid_n = 500, mp_max_iter = 50)
umi$curve
```

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